

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour

Total Marks:30

Annexure A

Resolution Algorithm Implementation

In

Code of Conduct:

I **Nethra P(192511012)** _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Nethra P

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.

2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

QUESTION 1 — Rain and Wet Ground

Given Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

Let us define two propositional symbols to represent the atomic statements involved in this problem:

- R = "It rains"
- W = "The ground is wet"

Using the above symbols, the knowledge base and the goal are represented as:

Premise 1 (Implication): $R \rightarrow W$

Premise 2 (Fact): R

Goal: W

Step 2: Conversion to Conjunctive Normal Form (CNF)

An implication of the form $P \rightarrow Q$ is logically equivalent to $\neg P \vee Q$ (this equivalence follows from the material implication rule). Applying this rule converts every conditional statement into a disjunctive clause suitable for resolution. Facts that are already atomic propositions require no conversion, as a single literal is itself a valid clause in CNF.

Original Statement	CNF Clause
$R \rightarrow W$	$\neg R \vee W$
R (fact)	R

Step 3: Negation of the Goal

To apply proof by refutation (which underlies the Resolution Algorithm), the goal is negated and added to the knowledge base. If the resulting clause set is unsatisfiable — i.e., resolution derives the empty clause — the original, un-negated goal is proved to logically follow from the knowledge base.

$\neg W$

This negated goal is added to the set of clauses derived from the knowledge base. The Resolution Algorithm will now attempt to derive a contradiction (the empty clause) from this combined clause set — if it succeeds, the original goal is proved true.

Step 4: Applying the Resolution Algorithm

The complete clause set available for resolution is:

C1: $\neg R \vee W$

C2: R

C3: $\neg W$ (negated goal)

Resolution is applied by finding two clauses that contain complementary literals (a literal and its negation), removing that complementary pair, and forming a new clause — the resolvent — from the remaining literals of both parent clauses. This process is repeated until either the empty clause ($\square$) is generated, proving the goal, or no further resolution is possible.

Step	Clauses Resolved	Resolvent Derived	Remark
1	C1 ($\neg R \vee W$) and C3 ($\neg W$)	$\neg R$	Literals W and $\neg W$ are complementary; they cancel, leaving $\neg R$.
2	Resolvent $\neg R$ and C2 (R)	$\square$ (Empty Clause)	Literals R and $\neg R$ are complementary; they cancel, leaving no literals — the empty clause.

Step 5: Empty Clause and Conclusion

The derivation of the empty clause ($\square$) indicates that the clause set $\{\neg R \vee W, R, \neg W\}$ is unsatisfiable — that is, these three clauses can never all be simultaneously true.

Since $\neg W$ (the negated goal) was one of the clauses contributing to this contradiction, it follows that $\neg W$ cannot be true whenever the knowledge base ($R \rightarrow W$ and R) is true. Therefore, the original goal, W ("the ground is wet"), must be true whenever the knowledge base holds.

Final Conclusion: The Resolution Algorithm successfully derives the empty clause ($\square$) in two steps, thereby proving that "the ground is wet" is a valid logical consequence of the given knowledge base.

QUESTION 2 — Student Assignment Submission

Given Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

Let the following propositional symbols represent the relevant statements about Rahul:

- S = "Rahul submits the assignment"
- M = "Rahul receives internal marks"

Using the above symbols, the knowledge base and the goal are represented as:

Premise 1 (Implication): $S \rightarrow M$

Premise 2 (Fact): S

Goal: M

Step 2: Conversion to Conjunctive Normal Form (CNF)

As before, the implication $S \rightarrow M$ is rewritten using the equivalence $P \rightarrow Q \equiv \neg P \vee Q$. The fact S remains a single-literal clause.

Original Statement	CNF Clause
$S \rightarrow M$	$\neg S \vee M$
S (fact)	S

Step 3: Negation of the Goal

The goal M is negated and appended to the clause set. If resolution on this expanded set yields a contradiction, the original goal M is proved.

$\neg M$

This negated goal is added to the set of clauses derived from the knowledge base. The Resolution Algorithm will now attempt to derive a contradiction (the empty clause) from this combined clause set — if it succeeds, the original goal is proved true.

Step 4: Applying the Resolution Algorithm

The complete clause set available for resolution is:

C1: $\neg S \vee M$

C2: S

C3: $\neg M$ (negated goal)

The algorithm searches for complementary literal pairs across the clause set and resolves them iteratively:

Step	Clauses Resolved	Resolvent Derived	Remark
1	C1 ($\neg S \vee M$) and C3 ($\neg M$)	$\neg S$	M and $\neg M$ cancel out, leaving $\neg S$.
2	Resolvent $\neg S$ and C2 (S)	$\square$ (Empty Clause)	S and $\neg S$ cancel out, leaving the empty clause.

Step 5: Empty Clause and Conclusion

The clause set $\{\neg S \vee M, S, \neg M\}$ is shown to be unsatisfiable, since resolution reduces it to the empty clause ($\square$) in two steps.

Because $\neg M$ cannot coexist with the knowledge base without producing a contradiction, the negation of the goal is false whenever the premises hold — which means the goal M itself must be true.

Final Conclusion: Since Rahul submitted the assignment (S is true) and submission implies receiving internal marks ($S \rightarrow M$), the Resolution Algorithm proves in two steps that Rahul receives internal marks (M) is a valid logical conclusion.

QUESTION 3 — Library Membership

Given Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

Let the propositional symbols be defined as follows:

- L = "Priya is a library member"
- B = "Priya can borrow books"

Using the above symbols, the knowledge base and the goal are represented as:

Premise 1 (Implication): $L \rightarrow B$

Premise 2 (Fact): L

Goal: B

Step 2: Conversion to Conjunctive Normal Form (CNF)

The implication is converted into clausal form using $P \rightarrow Q \equiv \neg P \vee Q$. Facts remain single-literal clauses. Both resulting clauses are already in CNF, as CNF requires a conjunction of clauses, each of which is a disjunction of literals — a single clause trivially satisfies this form.

Original Statement	CNF Clause
$L \rightarrow B$	$\neg L \vee B$
L (fact)	L

Step 3: Negation of the Goal

To construct a refutation proof, the goal B is negated and added to the clause set derived from the knowledge base:

$\neg B$

This negated goal is added to the set of clauses derived from the knowledge base. The Resolution Algorithm will now attempt to derive a contradiction (the empty clause) from this combined clause set — if it succeeds, the original goal is proved true.

Step 4: Applying the Resolution Algorithm

The complete clause set available for resolution is:

C1: $\neg L \vee B$

C2: L

C3: $\neg B$ (negated goal)

Resolution proceeds by repeatedly identifying and cancelling complementary literal pairs until no new clauses can be generated, or until the empty clause is produced:

Step	Clauses Resolved	Resolvent Derived	Remark
1	C1 ($\neg L \vee B$) and C3 ($\neg B$)	$\neg L$	B and $\neg B$ are complementary; they cancel, leaving $\neg L$.
2	Resolvent $\neg L$ and C2 (L)	$\square$ (Empty Clause)	L and $\neg L$ are complementary; they cancel, leaving the empty clause. No further clauses can be generated.

Step 5: Empty Clause and Conclusion

Since the empty clause ($\square$) is reached, the combined clause set $\{\neg L \vee B, L, \neg B\}$ is proved unsatisfiable.

As no further resolution is possible beyond this point (the algorithm terminates upon deriving $\square$), the process confirms that the assumption $\neg B$ leads directly to contradiction, and therefore B must hold under the given knowledge base.

Final Conclusion: The Resolution Algorithm terminates with the empty clause ($\square$) after two resolution steps, proving that Priya can borrow books (B) is logically entailed by the knowledge base.

QUESTION 4 — Placement Eligibility

Given Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

Define the propositional symbols relevant to Arun's case:

- A = "Arun clears the aptitude test"
- E = "Arun is eligible for placement"

Using the above symbols, the knowledge base and the goal are represented as:

Premise 1 (Implication): $A \rightarrow E$

Premise 2 (Fact): A

Goal: E

Step 2: Conversion to Conjunctive Normal Form (CNF)

The implication $A \rightarrow E$ is converted to clausal form using the standard equivalence $P \rightarrow Q \equiv \neg P \vee Q$, giving a two-literal clause. The fact A is already a single-literal clause.

Original Statement	CNF Clause
$A \rightarrow E$	$\neg A \vee E$
A (fact)	A

Step 3: Negation of the Goal

For the refutation proof, the goal E is negated and included in the working clause set:

$\neg E$

This negated goal is added to the set of clauses derived from the knowledge base. The Resolution Algorithm will now attempt to derive a contradiction (the empty clause) from this combined clause set — if it succeeds, the original goal is proved true.

Step 4: Applying the Resolution Algorithm

The complete clause set available for resolution is:

C1: $\neg A \vee E$

C2: A

C3: $\neg E$ (negated goal)

The Resolution Algorithm is applied step by step below, resolving pairs of clauses that contain complementary literals:

Step	Clauses Resolved	Resolvent Derived	Remark
1	C1 ($\neg A \vee E$) and C3 ($\neg E$)	$\neg A$	E and $\neg E$ cancel; the remaining

Step	Clauses Resolved	Resolvent Derived	Remark
			literal from C1 is $\neg A$.
2	Resolvent $\neg A$ and C2 (A)	$\square$ (Empty Clause)	A and $\neg A$ cancel; no literals remain, producing the empty clause.

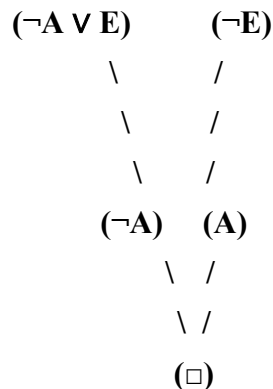
Step 5: Empty Clause and Conclusion

The empty clause ($\square$) is derived after two resolution steps, proving that the clause set $\{\neg A \vee E, A, \neg E\}$ is unsatisfiable.

Because the contradiction arises directly from assuming $\neg E$, the negated goal cannot be true alongside the knowledge base — hence the original goal E must be true.

Resolution Tree

The resolution tree below illustrates how the clauses combine, level by level, until the empty clause is reached at the root:



Final Conclusion: Following the resolution tree from the leaves ($\neg A \vee E$, A, and $\neg E$) down to the root ($\square$), the Resolution Algorithm proves that Arun is eligible for placement (E) is a valid logical consequence of clearing the aptitude test.

QUESTION 5 — Access Control System

Given Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access, using the Resolution Algorithm.

Step 1: Propositional Logic Representation

This problem involves a chain of two implications, so three propositional symbols are required:

- P = "The user enters the correct password"
- A = "The user is authenticated"
- G = "The user is granted access"

Using the above symbols, the knowledge base and the goal are represented as:

Premise 1 (Implication): $P \rightarrow A$

Premise 2 (Implication): $A \rightarrow G$

Premise 3 (Fact): P

Goal: G

Step 2: Conversion to Conjunctive Normal Form (CNF)

Each implication is individually converted to clausal form using $P \rightarrow Q \equiv \neg P \vee Q$. This yields two two-literal clauses, plus the single-literal fact clause. Because the two implications share the common term A , they form a logical chain ($P \rightarrow A \rightarrow G$) that resolution can traverse literal by literal.

Original Statement	CNF Clause
$P \rightarrow A$	$\neg P \vee A$
$A \rightarrow G$	$\neg A \vee G$
P (fact)	P

Step 3: Negation of the Goal

The goal G is negated and added to the clause set derived from the three premises, forming the full set on which refutation resolution is performed:

$\neg G$

This negated goal is added to the set of clauses derived from the knowledge base. The Resolution Algorithm will now attempt to derive a contradiction (the empty clause) from this combined clause set — if it succeeds, the original goal is proved true.

Step 4: Applying the Resolution Algorithm

The complete clause set available for resolution is:

C1: $\neg P \vee A$

C2: $\neg A \vee G$

C3: P

C4: $\neg G$ (negated goal)

Because this knowledge base involves a two-step implication chain, resolution must be applied across three successive steps to reach the empty clause. Each step cancels one complementary literal pair:

Step	Clauses Resolved	Resolvent Derived	Remark
1	C2 ($\neg A \vee G$) and C4 ($\neg G$)	$\neg A$	G and $\neg G$ are complementary; they cancel, leaving $\neg A$.
2	Resolvent $\neg A$ and C1 ($\neg P \vee A$)	$\neg P$	A and $\neg A$ are complementary; they cancel, leaving $\neg P$.
3	Resolvent $\neg P$ and C3 (P)	$\square$ (Empty Clause)	P and $\neg P$ are complementary; they cancel, leaving no literals — the empty clause.

Step 5: Empty Clause and Conclusion

The empty clause ($\square$) is derived after three resolution steps, confirming that the clause set $\{\neg P \vee A, \neg A \vee G, P, \neg G\}$ is unsatisfiable.

Step 1 uses the second implication ($A \rightarrow G$) to eliminate G, reducing the negated goal to $\neg A$. Step 2 uses the first implication ($P \rightarrow A$) to eliminate A, reducing the assumption further to $\neg P$. Step 3 uses the known fact P to eliminate $\neg P$ entirely, yielding the empty clause and completing the contradiction.

This three-step chain mirrors the logical chaining of the original premises: possession of the correct password (P) leads to authentication (A), which in turn leads to granted access (G) — a direct application of hypothetical syllogism realised through resolution.

Final Conclusion: By resolving the negated goal successively against the two implication clauses and finally against the fact P, the Resolution Algorithm derives the empty clause ($\square$) in three steps, proving that the user is granted access (G) is a valid logical consequence of entering the correct password.

Summary of Results

The table below summarises the outcome of applying the Resolution Algorithm to all five questions in this assessment. In every case, the empty clause ($\square$) was successfully derived, confirming that each stated goal is a valid logical consequence of its corresponding knowledge base.

Q. No.	Symbols Used	Resolution Steps	Result
1	R, W	2	Goal proved — $\square$ derived
2	S, M	2	Goal proved — $\square$ derived

Q. No.	Symbols Used	Resolution Steps	Result
3	L, B	2	Goal proved — $\square$ derived
4	A, E	2	Goal proved — $\square$ derived
5	P, A, G	3	Goal proved — $\square$ derived

In each question, the knowledge base consisted of one or more implications combined with known facts. Converting implications to clausal form ($P \rightarrow Q \equiv \neg P \vee Q$) allowed every premise to be expressed as a disjunction of literals suitable for resolution. Negating the goal and adding it to the clause set enabled a proof by refutation: successive resolution steps eliminated complementary literal pairs until no literals remained, yielding the empty clause ($\square$) and thereby confirming that the original (un-negated) goal follows necessarily from the knowledge base. Question 5 required one additional resolution step compared to the others because its knowledge base involved a two-link implication chain ($P \rightarrow A \rightarrow G$) rather than a single implication, illustrating how the Resolution Algorithm naturally generalises to longer chains of reasoning.